General Biology 1

First Semester

Supplementary Materials

Module 4

Prokaryotic vs Eukaryotic Cell

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What I Know

Directions: Read each situation carefully and choose the best answer.

1. Which structure is found in eukaryotic cells but not in prokaryotic cells?
   a. Cell membrane b.Cytoplasm
   c. Nucleus d. Ribosome
2. Which group of organisms is made up of prokaryotic cells?
   a. Animals b. Plants
   c. Bacteria d. Fungi
3. What is the main difference between prokaryotic and eukaryotic cells?
   a. Prokaryotic cells have a nucleus
   b. Eukaryotic cells have membrane-bound organelles
   c. Prokaryotic cells are larger than eukaryotic cells
   d. Eukaryotic cells do not contain DNA
4. Which of the following organisms is most likely made of eukaryotic cells?
   a. Bacterium b. Mushroom
   c. Cyanobacteria d. Archaea
5. Why is cell structure important in the classification of life forms?
   a. It determines the color of organisms
   b. It helps scientists group organisms based on similarities
   c. It shows the habitat of organisms
   d. It determines how fast organisms move

What I Need to Know

This supplementary module is designed to help you understand the difference between **Prokaryotic and Eukaryotic Cell**. In this lesson, you will explore how cell complexity contributes to diversity in organisms and in classifying life forms.

Lesson 4 Prokaryotic vs Eukaryotic

All living organisms are made of cells. Cells are classified as prokaryotic or eukaryotic. Understanding the differences between prokaryotic and eukaryotic cells helps explain the diversity and classification of living things.

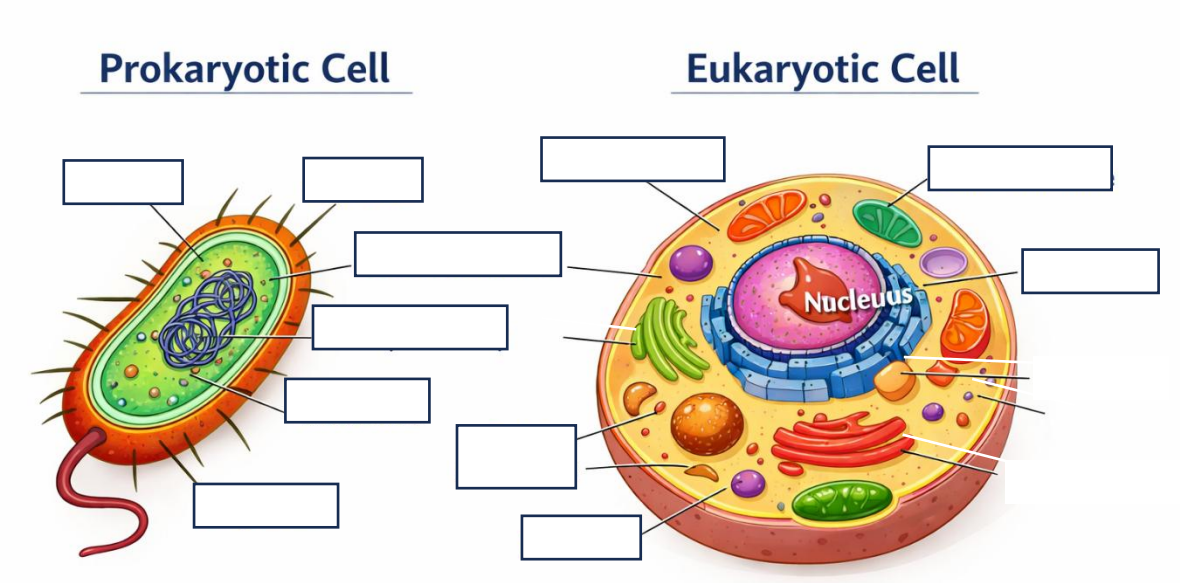
After going through this supplementary module, you are expected to:

1. differentiate prokaryotic and eukaryotic cells using diagrams and descriptions;
2. explain why cell complexity contributes to diversity in organisms; and
3. connect cell structure to the classification of life forms.

What's In

Activity 1: "Battle of the Cells: Simple vs. Complex"

Directions: Identify the parts of Prokaryotic and Eukaryotic Cells.
NOTE: There are organelle/s that are present in both cells. You can use the organelle twice.



Guide Questions:

1. Which is more complex, Prokaryotes or Eukaryotes? _____
2. Which organelle/s are present in both Prokaryotes and Eukaryotes? _____
3. If the DNA in Eukaryotes is found in the nucleus, how about in Prokaryotes? _____

- Nucleus
- Ribosomes
- Flagellum
- Cell Membrane
- Golgi Apparatus
- Nucleoid
- Pili
- Cell Wall

What's New

Activity 2: "Cell Complexity and Life Diversity Discussion Game"

Directions: Determine each scenario with either simple organisms (prokaryotes) or complex organisms (eukaryotes) and explain your reasoning.

Scenario	Eukaryotes or Prokaryotes	Reason
An organism reproduces quickly in a nutrient-rich environment. It has no nucleus and no membrane-bound organelles.		

This organism has mitochondria that generate large amounts of energy for movement and body regulation. It has a nucleus and several specialized organelles.		
A tiny single-celled organism performs photosynthesis. It lacks a nucleus but has chlorophyll inside its cytoplasm.		
A multicellular organism has chloroplasts, vacuoles, a cell wall, and a nucleus. It grows tall and performs photosynthesis efficiently.		
This organism is microscopic, has a cell wall, no nucleus, and reproduces rapidly, causing infections.		
A single-celled organism has a nucleus, mitochondria, and specialized organelles for moving and feeding.		
A tiny organism thrives in hot springs and acidic lakes, reproducing quickly with minimal internal structures.		
This cell contains mitochondria, a nucleus, ribosomes, and lysosomes. It supports rapid contraction and energy-intensive activities.		
A simple organism forms colonies, has a cell wall, plasma membrane, cytoplasm, and DNA in the nucleoid region, but no true nucleus.		
A unicellular organism has a nucleus, mitochondria, and endoplasmic reticulum. It		

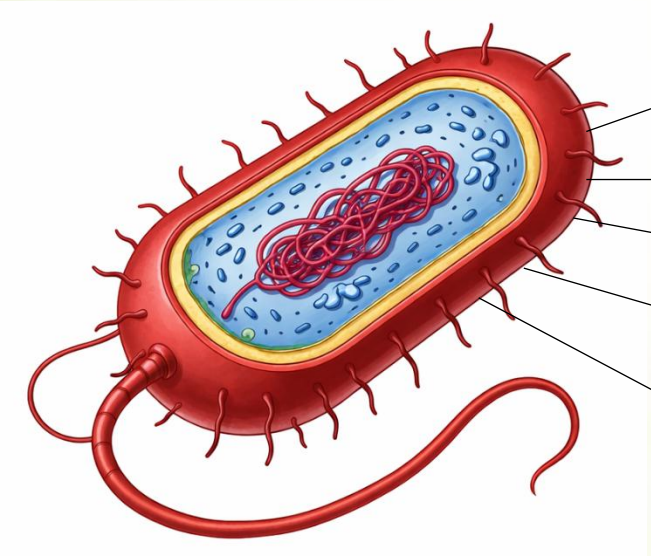
performs fermentation and produces energy efficiently.		

What's More

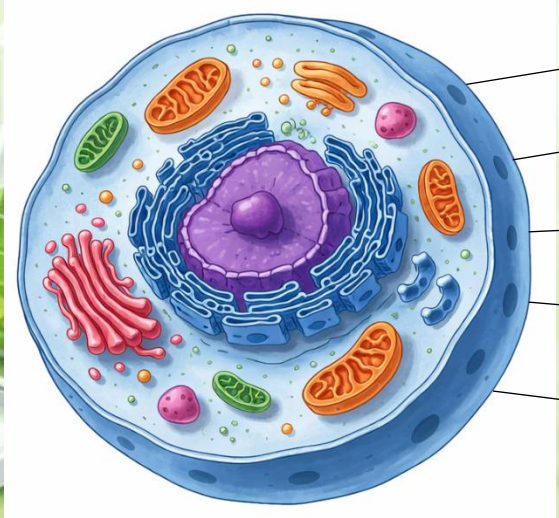
Activity 3: "Cell Quest: Unlocking Life's Classification"

Directions: Sort the organism cards by cutting them and pasting them into the Prokaryote or Eukaryote diagram.

Organisms



Organisms



ORGANISMS CARD

Amoeba	**Human Cell**
Streptococcus	**Blue-green-algae**
Paramecium	**Plant Cell**
E. coli	**Halophiles**
Archaea	**Yeast**

Guide Questions:

1. Why did you classify this organism as prokaryotic or eukaryotic?

   _____

   _____

2. What evidence supports your decision?

   _____

   _____

3. Which organisms were hardest to classify? Why?

   _____

   _____

All prokaryotic cells are structurally simple. They can be divided into two main groups, called the domain Bacteria and domain Archaea. **Bacteria** are well known because they can cause serious diseases, such as tuberculosis, tetanus, and throat infections. Many species of bacteria are important to the environment because they contribute to ecological cycles and decompose remains of dead organisms. They can also be used for manufacturing some products such as medicines and food. Unlike bacteria, **Archaea**, also known as archaebacteria, can live on extreme environments such as hypothermal vents and acidic or basic waters. Some of them can live in digestive tracts of animals and even on petroleum deposits underground. Archaea that can survive under extreme condition are called **extremophiles**.

Structures of a Prokaryotic Cell

1. **Cell Membrane** is a phospholipid bilayer with embedded proteins which regulates the entrance and exit of substances into and out of the cytoplasm.
2. **Mesosomes** are the extension of cell membrane used for attachment of enzymes that carry on metabolic activities.
3. **Cell wall** maintains the shape of the cell. The cell wall of bacteria contains peptidoglycan which is a complex molecule containing a unique amino disaccharide and peptide fragments.
4. **Cytoplasm** is a semifluid solution composed of water and inorganic and organic molecules encased by the plasma membrane.
5. **Glycocalyx** is a layer of polysaccharides lying outside the cell wall. It functions as a receptor and protection to the cell.
6. **Nucleoid** is a region where DNA is located. The DNA is coiled up into a single chromosome. It also contains extra-chromosomal plasmids.
7. **Sex Pilus** is a rigid tubular structure used by bacteria to pass DNA from cell to cell.
8. **Flagellum** helps in the movement of the cell. It has a filament, a hook, and a basal body.
9. **Fimbriae** are small, bristle-like fibers that sprout from the cell surface. They help- in the attachment of bacteria to the surface.
10. **Inclusion body** serves as storage of various substances such as carbohydrates, glycogen, phosphate, and fats which can be used when needed.
11. **Ribosome** is a tiny particle where protein synthesis takes place.
12. **Endospore** helps the cell in survive under harsh conditions.

A **eukaryotic cell** is a type of cell found in plants, animals, protists, and fungi. Unlike a prokaryotic cell-, it has a membrane-bounded nucleus which houses its DNA. Eukaryotic cells contain different organelles which perform different functions in the cell. Eukaryotes can be single-celled, colonial, or multicellular.

Structures of Eukaryotic Cells

1. **Nucleus** is where the genetic material, deoxyribonucleic Acid (DNA), is stored. It helps in the production of proteins and ribosomes.
2. **Cytoplasm** is a semifluid solution composed of water, inorganic and organic molecules that are encased by the cell membrane. It is

where all organelles are located and most of cellular activities take place.

3. **Mitochondrion** is responsible for production of adenosine triphosphate (ATP), the energy source of cellular processes, and where cellular respiration happens.

4. **Chloroplast** can only be found in plants and algae. It is responsible for converting energy to chemical energy through the process of photosynthesis.

5. **Golgi apparatus** helps in the packaging and sorting of modified cell products of proteins and lipids.

6. **Lysosome** is a vesicle that digests intracellular molecules and even cell parts.

7. **Vacuole** is responsible for engulfing materials through the process of endocytosis.

8. **Endoplasmic reticulum** transports lipids, proteins, and other materials through the cell. The two types of endoplasmic reticulum are smooth endoplasmic reticulum and rough endoplasmic reticulum.

9. **Cilia and flagella** are hair-like projections that enables the cell to move. Cilia are much shorter than flagella, but they have similar construction.

10. **Cell/ Plasma membrane** is a phospholipid bilayer that regulates the entrance and exit of substances in the cell.

11. **Ribosome** is a particle that carry out protein synthesis.

12. **Cytoskeleton** is responsible in maintaining the shape and framework of the cell. It is also responsible in assisting movement of cell organelles. It is composed of actin filaments, intermediate filaments, and microtubules.

- **Actin filaments** are protein fibers that play a role in the movement of the cell and the organelles.
- **Intermediate filaments** are protein fibers that provide support and strength to the cell.
- **Microtubules** are cylinders of protein molecules present in cytoplasm, centrioles, cilia, and flagella.

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What I've Learned

Directions: Encircle the letter of the correct answer.

1. A scientist discovers a unicellular organism that has ribosomes but no nucleus or membrane-bound organelles. What is the best classification?
   a. Eukaryote because it has ribosomes
   b. Prokaryote because it lacks a nucleus
   c. Eukaryote because it is unicellular
   d. Virus because it is simple
2. An organism is found in a hot spring and can survive extreme temperatures. It has no nucleus. Which group does it most likely belong to?
   a. Fungi
   b. Animalia
   c. Archaea
   d. Protista
3. Which situation best explains why eukaryotic cells are more complex than prokaryotic cells?
   a. They are always larger in size
   b. They have specialized organelles that perform different functions
   c. They reproduce faster
   d. They lack a cell membrane
4. A student claims: "All unicellular organisms are prokaryotic." Which response best evaluates this statement?
   a. Correct, because unicellular organisms are simple
   b. Incorrect, because some unicellular organisms have a nucleus
   c. Correct, because prokaryotes are always unicellular
   d. Incorrect, because all eukaryotes are multicellular
5. Why can bacteria reproduce more quickly than most eukaryotic cells?
   a. They have more DNA
   b. They lack complex organelles, making processes faster
   c. They have a nucleus
   d. They are multicellular